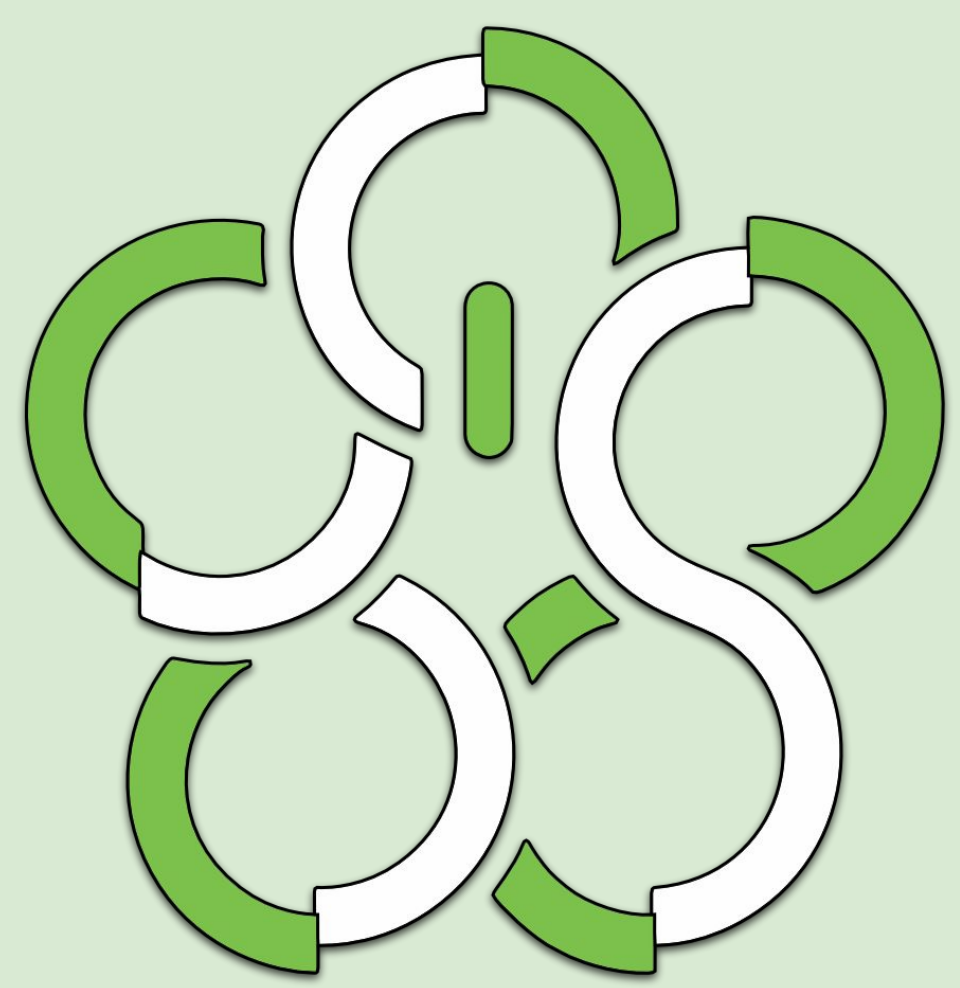




Benchmarking Vision-Language Models on Cross-View Camera Orientation Estimation



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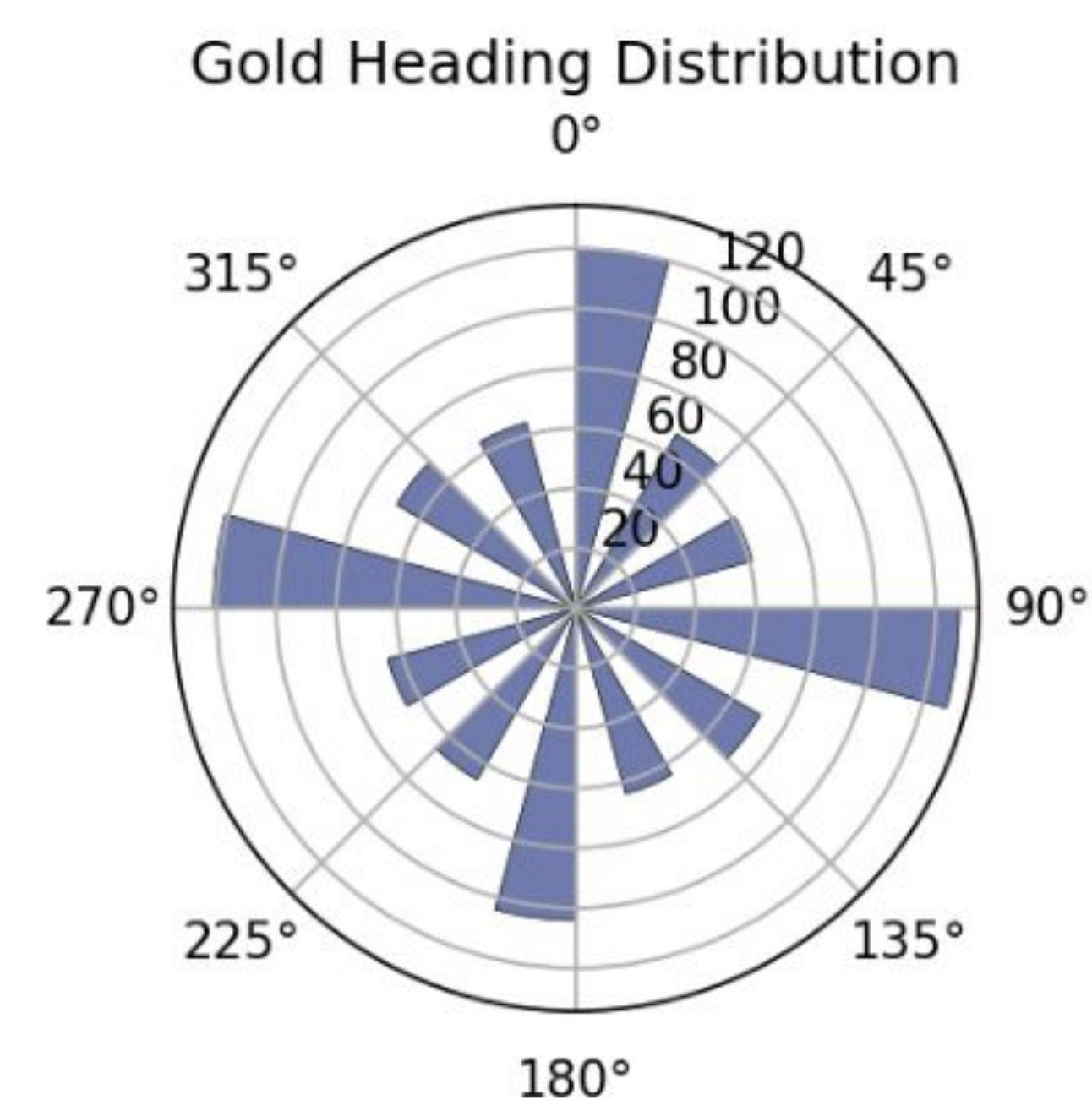
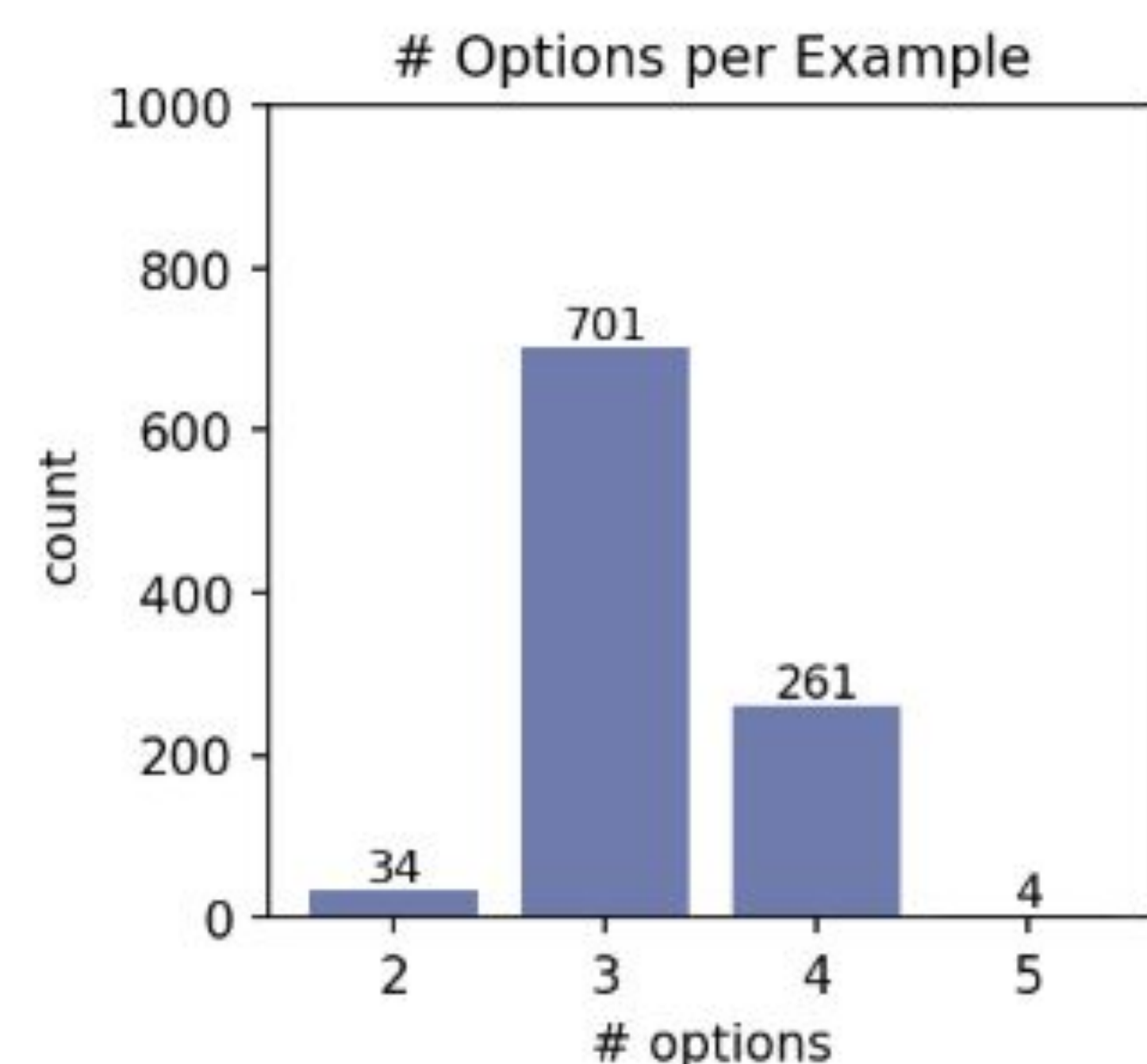
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OVERVIEW

- Accurate spatial orientation is essential for GPS-denied embodied AI.
- Cross-view geospatial reasoning is underexplored in VLMs.
- Existing benchmarks cover semantic interpretation and global localization, but not fine-grained cross-view orientation estimation.
- This project expands the original Map 2 Street View (M2SV) benchmark with 3 new task variations.

DATASET

- Aerial and ground-level imagery was collected via Google Maps and Street View Static APIs across 32 globally distributed cities, with intersection nodes sampled from OpenStreetMap road networks



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TASK DESCRIPTION

Model must reason about the egocentric street-view images in reference to the allocentric aerial images, while camera is known to be fixed at the center and maps are North-aligned.

(A) Allocentric Selection: Select labeled vector on the aerial image that matches the direction of the street view image. (original M2SV task)



Output: "C"

(B) Egocentric Selection: Select the street-view image that matches the labeled vector on the aerial image.



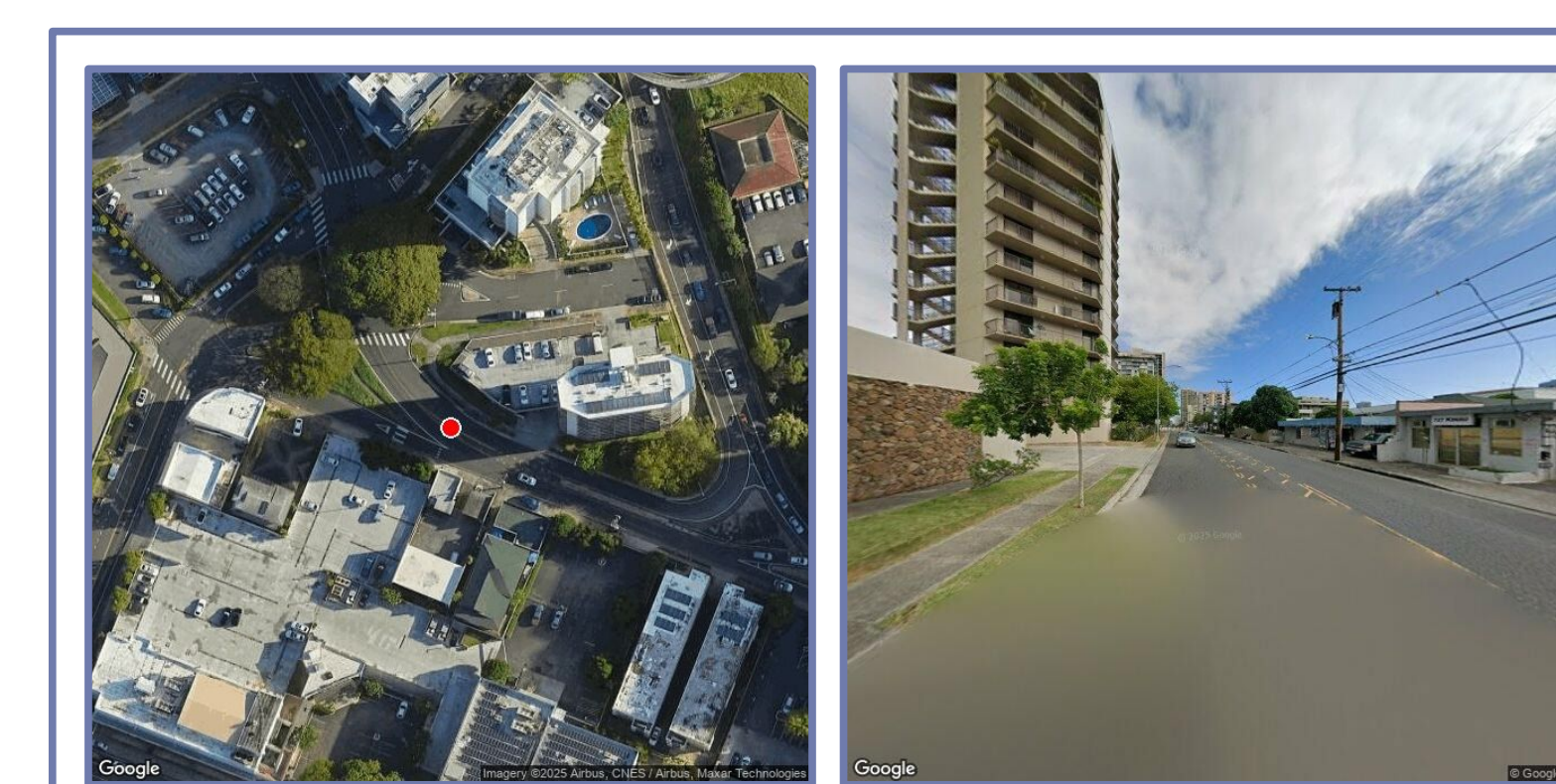
Output: "3"

(C) Multi-View Matching: Give the one-to-one matching between the sets of street-view images and candidate directions.



Output: "A:2, B:3, C:1"

(D) Angle Estimation: Give the raw azimuth angle without visual hints.



Output: "120"

EXPERIMENTS

Human Baseline Evaluation

- Two annotators, 100 examples per dataset.
- Tasks are intuitive for humans.

Zero-Shot Prompt Evaluation

- All models evaluated zero-shot, no fine-tuning or few-shot examples, 1000 examples per dataset.
- Goal:** establish a baseline for off-the-shelf VLM geospatial reasoning ability.

Metrics

- Accuracy:** # of correctly labeled multiple choice answers.
- Mean Absolute Error (MAE):** Average absolute angular difference between pred and gold.
- Accuracy @ k°:** # of answers correctly labeled within $\pm k^\circ$ threshold.

RESULTS

Model	Allocentric (A)	Egocentric (B)	Multi-View (C)	Avg.
Human baseline	81.0%	88.0%	75.5%	81.5%
Random baseline	31.7%	31.7%	31.7%	31.7%
Gemini 2.5 Pro	45.6%	49.4%	46.8%	47.3%
Gemini 2.5 Flash	41.3%	42.1%	34.2%	39.2%
Qwen3 VL 8B Thinking	34.4%	33.8%	18.5%	28.9%
Ministral 3 8B 2512	31.1%	35.4%	17.0%	27.8%
Ministral 3 14B 2512	33.1%	33.8%	16.6%	27.8%
Gemma 3 12B (it)	31.6%	33.7%	15.0%	26.8%
Qwen3 VL 8B Instruct	35.5%	25.0%	11.8%	24.1%
Gemma 3 4B (it)	30.2%	26.4%	11.5%	22.7%

Table 1. Overview of model accuracy across allocentric, egocentric, and multi-view tasks.

Model	MAE	Acc @ 15°	Acc @ 30°	Acc @ 45°
Random baseline	90°	8.3%	16.7%	25.0%
Gemini 2.5 Pro	84°	22.3%	30.2%	34.1%
Gemini 2.5 Flash	83°	18.2%	26.4%	32.9%

Table 2. Model mean absolute error and accuracy within 15°, 30°, and 45° margins on the angle estimation task.

KEY FINDINGS

- Open-weight models at the 8-14B scale struggle to maintain spatial representations across multiple images in a single inference pass.
- Instruction-tuned training may suppress the open-ended spatial reasoning these tasks demand.
- Multi-view matching task requires reasoning in both allocentric and egocentric directions → Weaker models show steeper drops in performance.
- Ground-level street-view images closely resemble the natural image distributions seen during pretraining → Higher scores on egocentric task.